

63. Peritoneal Dialysis · assets 1-7 / 7

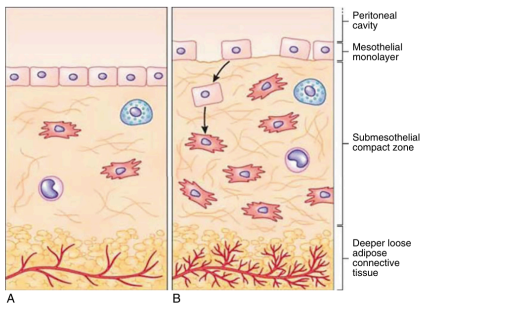


Fig. 63.1 Changes in the peritoneal membrane over time. Initially (A), the peritoneal membrane baseline architecture. After ongoing treatment with dialysis (B), the mesothelial monolayer undergoes changes, leading to an epithelial-to-mesenchymal transition of the cells toward a myofibroblastic morphology. The submesothelial compact zone thickens due to increased extracellular matrix deposition. Deeper in the adipose-rich connective tissue, the vascular network is subject to neovascularization and escalating vasculopathy, marked by hyalinization of arterioles and venules, luminal enlargement, and obliteration. (From Nessim SJ, Perl J, Bargman JM. The renin-angiotensin-aldosterone system in peritoneal dialysis: is what is good for the kidney also good for the peritoneum? *Kidney Int.* 2010;78(1):23–28.)

Fig_63_1

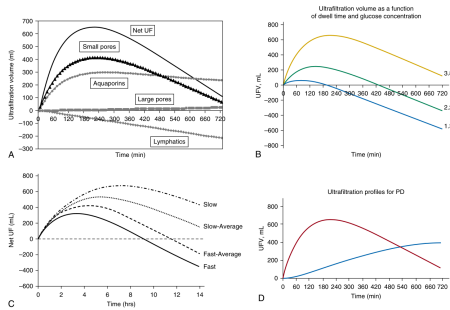


Fig. 63.2 (A) Computer model depicting ultrafiltration (UF) across the peritoneal membrane using a 3.86% glucose solution demonstrates that, following the dissipation of the osmotic gradient at 300 minutes, fluid reabsorption occurs via small pores and lymphatic pathways. Due to the correlation of solute transport with the area of small pores, membranes with higher transport capabilities absorb more fluid. (B) The ultrafiltration volume (UFV) over time is represented for different glucose concentrations: 3.86% (yellow), 2.27% (green), and 1.36% (blue), using a three-pore model simulation. (C) The graph shows a comparison between the ultrafiltration profile for 7.5% icodextrin (blue line) and a 3.86% glucose solution (red line), both simulated with the three-pore model. (D) Variations in UF are observed for different peritoneal membrane solute transfer rates using a 2.5, 2.5% dextrose solution. Patients with rapid solute transfer rates in their peritoneal membranes experience reduced maximum UF and a quicker onset of fluid reabsorption. (Adapted with permission from [A]^[28] [C]^[29] and [B] and [D]^[30])

Fig_63_2

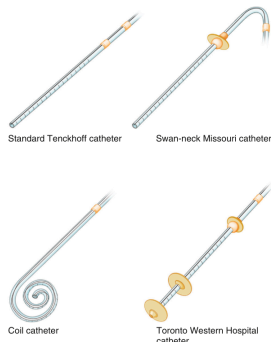


Fig. 63.3 Differences in the intraperitoneal and extraperitoneal segments between currently available peritoneal dialysis catheters used in clinical practice.

Fig_63_3

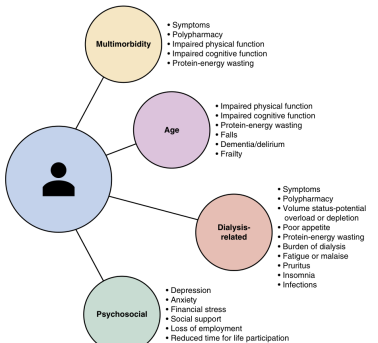


Fig. 63.4 Factors influencing outcomes for patients on peritoneal dialysis when using an individualized approach to care. (Adapted with modification from Brimble KS, Walker M, Margolis PJ, et al. Meta-analysis: peritoneal membrane transport, mortality, and technique failure in peritoneal dialysis. *J Am Soc Nephrol.* 2006;17(8):2591–2608.)

Fig_63_4

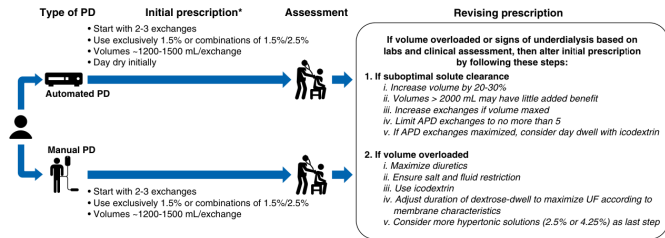


Fig. 63.5 Oversimplified approach of an incremental peritoneal dialysis prescriptive strategy. (© Bourne L, Auguste.)

Fig_63_5

Table 63.1 2022 ISPD Definitions of Peritonitis-Related Outcomes and Various Types of Peritonitis	
Term	Definition
Medical cure	Complete resolution of peritonitis together with KQME of the following complications: relapse/recurrence of peritonitis, catheter removal, transfer to HD for 300 days or death
Refractory	Peritonitis episode with persistently elevated days or persistent days after 14 days of appropriate antibiotic therapy
Recurrent	Peritonitis episode that occurs within 4 weeks of completion of therapy of a prior episode but with a different organism
Relapsing	Peritonitis episode that occurs within 4 weeks of completion of therapy of a prior episode with the same organism or one episode (culture negative) episode relapse (culture negative) followed by a specific organism or specific organism followed by culture negative
Repeat	Peritonitis episode that occurs more than 4 weeks after completion of therapy of a prior episode with the same organism
Peritonitis-associated catheter removal	Removal of PD catheter as part of the treatment of an active peritonitis episode
Peritonitis-associated death	Transfer from PD to HD for any period as part of the treatment for a peritonitis episode
Peritonitis-associated death	Death occurring within 30 days of peritonitis onset or death during hospitalization due to peritonitis
Peritonitis-associated hospitalization	Hospitalization precipitated by the occurrence of peritonitis for the purpose of peritonitis treatment delivery

Table_63_1

Table 63.2 Recommended Duration of Antimicrobial Treatment for Various Organisms Associated With Peritoneal Dialysis Peritonitis and Important Considerations

Organism	Treatment Duration	Important Considerations
CNST (Coagulase-negative Staphylococcus)	2 weeks	• Switch to vancomycin if methicillin resistant even if cell count improving on a cephalosporin
Staphylococcus aureus	3 weeks	• Consider adjunct rifampin
Streptococci	2 weeks	• Check oral flora and dental hygiene
Enterococci	3 weeks	• Consider CAPD if ampicillin sensitive in every exchange
Nonpseudomonal gram negative	3 weeks	• SPICE/ESBL consider longer treatment time, 2 antibiotics to reduce relapse risk
Corynebacterium diphtheroid	3 weeks	• Laboratories may not report sensitivities (may treat as contaminant)
Pseudomonas	3 weeks	• Switch to vancomycin
Stenotrophomonas	3 weeks	• Close examination of exit and tunnel site
Culture negative	2 weeks	• Use Sulfamethoxazole-Trimethoprim if sensitive to Stenotrophomonas
		• If cell count improving, can stop gram-negative coverage
		• If cell count not improving, consider unusual organisms (i.e., typical, atypical mycobacteria)
Polymicrobial	3 weeks	• If multiple enteric organisms, imaging to rule out surgical peritonitis
		• Consider addition of empiric anaerobic coverage

Table_63_2